

AMARNATH S. PATEL

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Scientific software — numerical methods, GPU acceleration, and verification you can actually check.

EDUCATION

University of Central Florida

B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor

Undergraduate Student 4.00 GPA

August 2025 – Expected May 2029

- **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Mathematical Methods for Physics
- **Computing:** Data Structures & Algorithms, Computer Architecture, Computer Logic Design, Object-Oriented Programming, Discrete Structures, Deep Learning

Florida Atlantic University

111 credit hours completed between ages 14 and 18, alongside high school

University coursework via FAU High School, ages 14–18 3.66 GPA

August 2021 – May 2025

SCIENTIFIC SOFTWARE

CELERIS — electromagnetic solver and metasurface design pipeline. C++23, CUDA, Python, CMake

2026

- Wrote a full rigorous coupled-wave analysis solver from scratch rather than wrapping an existing kernel: per-layer Fourier-basis eigenproblems, dense complex linear algebra, Li and Liu–Fan inverse-rule factorization, the 2D-vectorial $P \cdot Q$ formulation for bi-periodic structures, and a Redheffer scattering-matrix recursion chosen over the transfer-matrix form for numerical stability across many layers.
- Treated verification as a first-class feature: every layer is checked against closed-form physics, an independent method, or energy conservation — $1e-6$ against the transfer-matrix method, $1e-15$ on multilayer recursion, $5e-12$ on a degenerate-case reduction, Σ (diffraction efficiency) = 1.000000 — and the assembled solver is cross-validated against the grcwa and Stanford S^+ codes to $\approx 1e-7$, reproducing canonical published results (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented Rayleigh–Sommerfeld far-field propagation as a CUDA kernel and benchmarked it honestly — against the project’s own optimized 16-core CPU path, not against a naive baseline: $4.8\text{--}5.8\times$ at 92k elements, agreeing to machine precision, with automatic CPU fallback. Diagnosed the kernel as memory-bound rather than compute-bound from the way the speedup declined as the grid grew.
- Built the science on top of the kernel: material dispersion (Sellmeier and tabulated n, k), parameter sweeps, gradient-based (Adam) inverse design, Monte-Carlo tolerance analysis, a wavefront/MTF/Strehl analysis battery, and GDSII export. Python bindings and a native Qt desktop GUI over the C++ core. *Sole-author manuscript in preparation (Computer Physics Communications).*

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL

August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Built the lab’s [data-acquisition system](#): a Python stack that drives four instruments as one coordinated experiment across GPIB/VISA, RS-232 and GigE Vision, with in-loop optimization of a live objective and automatic rejection of saturated frames. Reduced a full measurement campaign from days of manual operation to a single unattended run.
- Implemented the reduction pipeline that turns raw interferograms into physics: FFT sideband isolation and demodulation, Butterworth filtering, joint nonlinear optimization over three coupled parameters, and decomposition onto an analytic mode basis (NumPy/SciPy). Characterized at $\approx 98\%$ reconstruction fidelity against simulated ground truth.
- Contribute mount control and pointing automation to [PolyOculus](#), coordinating a networked array of 8 telescopes as one instrument.

Undergraduate Research Assistant — UCF Physics Department

March 2026 – Present

Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Built [ESTELA](#) (Rust, Tauri 2), a research tool generating multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with export to four formats. In production use by external adopters outside the university; manuscript in preparation.
- Built an agentic generation–evaluation pipeline over the Azure OpenAI batch API that drafts and validates candidate problems against instructor criteria.

Other software

2023 – Present

- Systems and tooling in C, Rust and Go: [hush](#), an interactive shell in C; [limebar](#), a Wayland status bar in C; [constrict](#), a build system in Go written to replace make; [coat](#), a universal theme applier in Rust.
- Embedded Linux platform (Arch Linux ARM on a Raspberry Pi 5, gpsd, USB serial, WireGuard) behind live worldwide telemetry for a [four-passenger solar car](#) — 2nd in division, Lockheed Martin Award for engineering excellence.

TECHNICAL SKILLS

Numerical methods: dense and complex linear algebra, eigenvalue problems, FFT and signal processing, gradient-based optimization, Monte-Carlo methods, phase retrieval and modal decomposition, error analysis and convergence testing

High performance: CUDA kernel development and profiling, multicore CPU optimization, memory-bound versus compute-bound analysis, HPC cluster work

Languages and tools: C++ (C++23), CUDA, C, Python (NumPy, SciPy), Rust, Go; Linux, Git, CMake, Docker, Qt/PySide6, LaTeX/Typtst; GPIB/VISA, RS-232, GigE Vision, FITS, GDSII

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition)

2025

Lockheed Martin Award, “Highest Level of Engineering Excellence,” AEV Competition

2024

1st Place, Night Hacks Hackathon — UniUtils

2023